

S.2 Table: Functional traits for each bee species based on measurements, laboratory observations and literature source.

| Bee species            | Left wing length<br>(mm) | ITS<br>(mm) | Left posterior tibia length<br>(mm) | Nesting material<br>(own observation) | Tongue length<br>(mm) | Pollen transport<br>(reference)             |
|------------------------|--------------------------|-------------|-------------------------------------|---------------------------------------|-----------------------|---------------------------------------------|
| <i>A. vilhenae</i>     | 7.32                     | 3.03        | 1.78                                | resine and plants                     | 1.05                  | Abdomen_scopa<br>(Silveira & Almeida, 2002) |
| <i>C. trigonoides</i>  | 10.24                    | 4.19        | 5.19                                | sand and resin                        | 2.67                  | Leg_scopa<br>(Buchmann, 2004)               |
| <i>C. analis</i>       | 9.04                     | 4.02        | 4.8                                 | sand and resin                        | 2.44                  | Leg_scopa<br>(Buchmann, 2004)               |
| <i>E. cognata</i>      | 9.28                     | 3.62        | 4.59                                | resin                                 | 6.8                   | Corbicula<br>(Saleh & Ramírez, 2019)        |
| <i>E. despecta</i>     | 7.78                     | 3.37        | 4.31                                | resin                                 | 7.11                  | Corbicula<br>(Saleh & Ramírez, 2019)        |
| <i>E. modestior</i>    | 8.39                     | 4.18        | 4.12                                | resin                                 | 7.05                  | Corbicula<br>(Saleh & Ramírez, 2019)        |
| <i>E. towsendi</i>     | 8.25                     | 4.29        | 3.92                                | resin                                 | 7.26                  | Corbicula<br>(Saleh & Ramírez, 2019)        |
| <i>M. sejuncta</i>     | 8.22                     | 3.41        | 2.54                                | leaves                                | 1.48                  | Abdomen_scopa<br>(Silveira & Almeida, 2002) |
| <i>M. turpis</i>       | 9.27                     | 3.39        | 2.77                                | leaves                                | 1.87                  | Abdomen_scopa<br>(Silveira & Almeida, 2002) |
| <i>M. chrysosarus</i>  | 9.74                     | 3.66        | 2.96                                | leaves                                | 1.8                   | Abdomen_scopa<br>(Silveira & Almeida, 2002) |
| <i>M. brasiliensis</i> | 8.45                     | 3.05        | 2.7                                 | leaves                                | 1.27                  | Abdomen_scopa<br>(Silveira & Almeida, 2002) |
| <i>Tetrapedia sp1</i>  | 8.17                     | 2.14        | 3.89                                | sand and resin                        | 1.23                  | Leg_scopa<br>(Buchmann, 2004)               |
| <i>Tetrapedia sp2</i>  | 9.55                     | 2.52        | 3.13                                | sand and resin                        | 1.27                  | Leg_scopa<br>(Buchmann, 2004)               |
| <i>C. tarsata</i>      | 9.04                     | 3.89        | 3.81                                | sand and resin                        | 1.29                  | Leg_scopa<br>(Buchmann, 2004)               |

## References

- Buchmann, S.L. (2004) Aspects of Centridine biology (Centris spp.) importance for pollination, and use of *Xylocopa* spp. as greenhouse pollinators of tomatoes and other crops. *Solitary Bees: Conservation, Rearing and Management for Pollination* (Freitas, B.M. & Pereira, J.O.), pp.203–211, Imprensa Universitária.
- Saleh, N.W. & Ramírez, S.R. (2019) Sociality emerges from solitary behaviours and reproductive plasticity in the orchid bee *Euglossa dilemma*. *Proc. R. Soc.* **(286)**, 20190588.  
<https://doi.org/10.1098/rspb.2019.0588>
- Silveira, F.A. & Almeida, E.A.B. (2002) Abelhas Brasileiras: Sistemática e Identificação. Fundação Araucária.

